

Trainings ▾

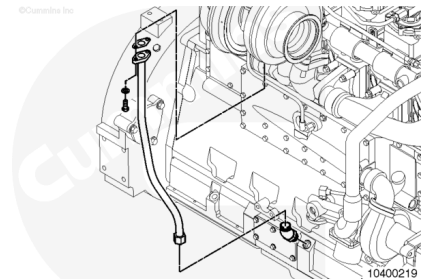
Remove

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



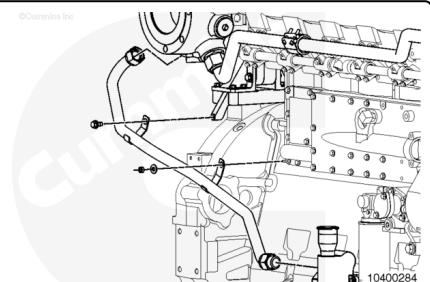
Disconnect the turbocharger oil drain tube from the male union.

Remove the two capscrews and lockwashers securing the turbocharger oil drain tube to the turbocharger. Discard the gasket.

Remove the male adapter elbow from the engine. Discard the gasket.

Marine Applications

Remove the two tube brackets and turbocharger oil return lines.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

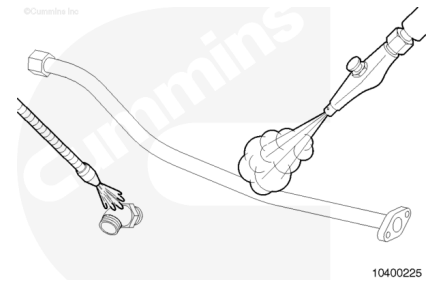
Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the turbocharger oil drain tube and male union adapter elbow.

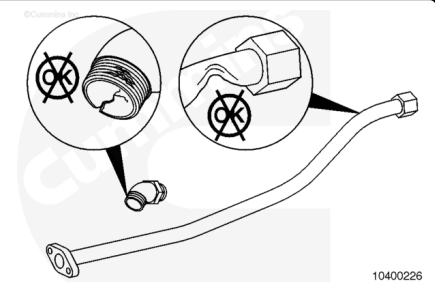
Dry with compressed air.



Inspect the turbocharger oil drain tube for any restrictions that would prevent the oil from free flowing.

Replace the turbocharger oil drain tube if kinks, other damage, or leakage is found.

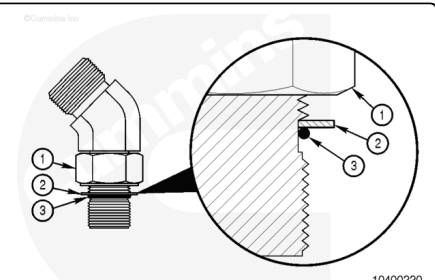
Inspect the male union adapter elbow for damage. Replace it if it is damaged, cracked, or has stripped threads.



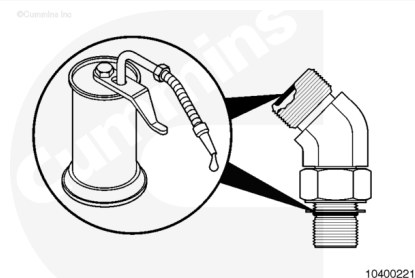
Install

Install the male adapter elbow by first positioning the locknut.

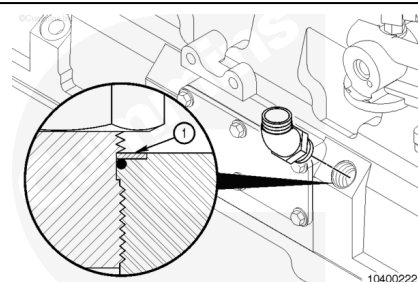
1. Locknut
2. Washer
3. O-ring



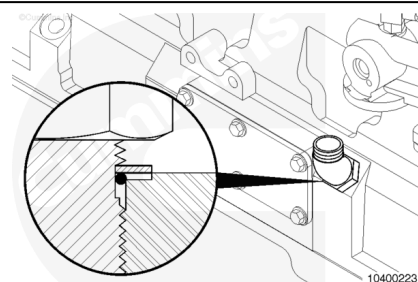
Install the new o-rings on the male adapter elbow and apply a light coating of clean engine oil to each o-ring.



Install the male adapter elbow in the engine port until the backup washer (1) contacts the port face.

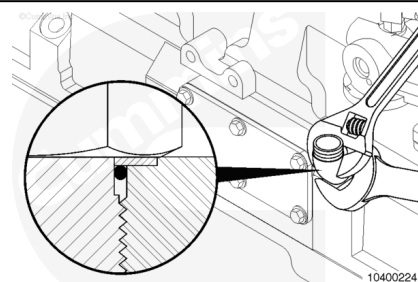


Position the male adapter elbow to align for connection to the turbocharger oil drain tube. Do **not** back off more than one full turn.



Use two wrenches to hold the male adapter elbow in the desired position and tighten the locknut.

Torque Value: 170 n•m [125 ft-lb]

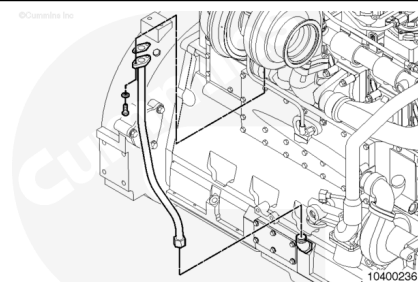


⚠ CAUTION ⚠

Do not use sealant or gasket adhesive on gasket as this can contaminate the oil system.

Install the new turbocharger oil drain gasket to the oil drain tube and attach it to the turbocharger with two lockwashers and capscrews.

Torque Value: 55 n•m [40 ft-lb]



Connect the turbocharger oil drain tube to the male adapter elbow on the engine.

Torque Value: 92 n•m [68 ft-lb]

Marine Applications

Install the turbocharger oil drain tube.

Torque Value:

Turbocharger Oil Drain Tube

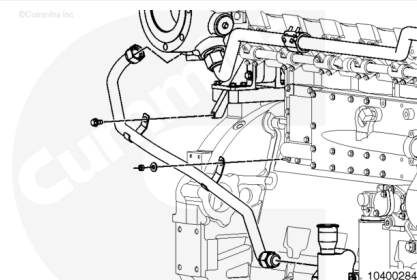
1. 43 to 47 n•m [32 to 35 ft-lb]

Install the capscrews for the brackets.

Torque Value:

Turbocharger Bracket Capscrews

1. 40 n•m [30 ft-lb]



Finishing Steps

Operate the engine and check for leaks.

